STATEMENT OF WORK (SOW)

Aviation Operations Center Moisture Intrusion Study

1.0 Project Objective

The objective of this study is to investigate water damage, mold growth, and suspected moisture intrusion at the Aviation Operations Center located at Fort Worth, Texas 76177, determine the root causes of the infiltration, and provide actionable recommendations to restore the building envelope's integrity and prevent future damage.

2.0 Scope of Services

The Consultant shall provide all labor, tools, equipment, and expertise to perform a comprehensive building envelope assessment, including:

- **Document Review:** Review construction documents, architectural drawings, previous repair reports, and maintenance records.
- **Visual Assessment:** Conduct a top-to-bottom inspection of the building exterior (roofing, cladding, glazing) and interior (walls, ceilings, subfloors) for signs of moisture, including water staining, efflorescence, peeling paint, or mold.
- **Non-Destructive Testing (NDT):** Utilize tools to detect hidden moisture, such as:
    - **Infrared Thermography (IR):** To identify temperature differentials suggesting moisture infiltration behind surfaces.
    - **Moisture Meters (Protimeter/Scanner):** To establish moisture mapping, quantifying moisture levels in materials (e.g., concrete, drywall, wood) compared to a "dry standard".
- **Destructive Testing/Exploratory Investigation:** Perform targeted, limited destructive testing (e.g., removing small sections of drywall, inspecting behind flashing) to confirm suspected leak sources, subject to client approval.
- **HVAC Assessment:** Review the mechanical system’s role in air pressure and humidity control to ensure moisture is not being introduced through condensation.

3.0 Investigative Procedures

The investigation will focus on the following potential moisture sources:

1. **Rainwater:** Penetration through walls, roofs, windows, and foundations.
2. **Groundwater:** Seepage through foundation walls or slabs (basements).
3. **Air Transportation:** Condensation of humid air migrating through wall assemblies.
4. **Vapor Diffusion:** Vapor moving through permeable materials.
5. **Capillary Action (Wicking):** Porous building materials, such as concrete or wood, pull liquid water up from the ground or wet areas against gravity.

Internal Sources of Moisture

1. **Condensation:** Exceeding the dew point and turning into liquid.
2. **Internal Leaks:** Broken pipes, leaking plumbing, or clogged HVAC condensate drains.

4.0 Deliverables

The Consultant shall submit a written Forensic Moisture Investigation Report containing:

1. **Executive Summary:** A concise summary of findings.
2. **Methodology:** Description of tools and techniques used (IR, meters, etc.).
3. **Findings & Moisture Maps:** Detailed maps indicating areas of elevated moisture.
4. **Root Cause Analysis:** Identification of why water is entering the building (e.g., failed sealant, missing flashing).
5. **Photographic Documentation:** Labeled photos of all inspection areas and identified defects.
6. **Remediation Recommendations:** Specific repair methods to correct the issues and prevent recurrence.

5.0 Project Schedule

- **Site Inspection:** To be determined

- **Draft Report Submittal:** To be determined
- **Final Report Submittal:** To be determined

6.0 Health and Safety

The contractor must follow safety protocols for working in potentially wet or mold-contaminated areas, including appropriate PPE. Any suspected microbial growth (mold) greater than 10 ft² must be reported to the Owner immediately